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From posthumanism to ethics of artificial intelligence

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Abstract
Posthumanism is one of the well-known and significant concepts in the present day. It impacted numerous contemporary fields like philosophy, literary theories, art, and culture for the last few decades. The movement has been concentrated around the technological development of present days due to industrial advancement in society and the current proliferated daily usage of technology. Posthumanism indicated a deconstruction of our radical conception of '*human*', and it further shifts our societal value alignment system to a novel dimension. The majority of our population is getting deeply involved in virtual reality in daily life. Sooner or later, we shall get a different conception of 'biological human being' through the advancement of artificial intelligence (AI) technology. If an automated artificial system could replace the human brain and repair any physical loss of our biological body, it will certainly become a journey towards immortality for scientists. However, we must analyze whether posthumanism will consider 'hybrid human beings' as moral agents, similar to biological humans. This is why, in the future, the relation between biological human beings and posthumans will play an active role in designing artificial moral agents. Whether the future posthumans would overpower biological humanity or both of them would work as peers to form a digital utopian society and create new dimensions of rationality is still a case of anticipation. Our aim in this paper is to critically analyze the authenticity of the posthuman cyborg as an agent, their relations with humans and the emergence of 'AI ethics'.

Keywords Posthumanism · Postmodernity · Artificial intelligence (AI) · Artificial moral agent · Ethics · Moral agent

1 Introduction

'Posthumanism' is one of the well-known and significant concepts in the present day. The first critical use of the term 'posthuman' can be found in the writings of Hassan in 1977.[1] The terminology denotes a conflicting approach toward radical anthropocentrism. Because the emergence of 'posthumanism' has modified the trajectory of human importance in world history. It emphasizes the importance of the existence of other species, nature. Therefore, 'posthumanism' holds a conflict toward anthropocentrism's hierarchical division, embedded in the conception of 'human'. Friedrich Wilhelm Nietzsche has used the same idea in describing his philosophical conception of '*übermensch*' or '*overman*', as the person who will be devoid of human limitations. He will be the next *Godly* person to the commons, who can navigate his peers to a comparatively free society.[2] Nietzsche has argued that his '*overman*' would possess some extraordinary qualities which are beyond the capabilities of any ordinary human being.[3]

The posthumanism movement has left many marks in contemporary history. It includes several variants like cultural posthumanism, critical posthumanism, philosophical posthumanism. The importance of nature, other species, and even the emergence of feminist movement were the most popular outcomes of philosophical posthumanism. Another significant trace can be found in the contemporary developments of automated industry and upsurge of AI systems. AI technology as a prospering field, is helping to achieve human immortality from their biological limitations.[4] In this context, Philosopher Nick Bostrom has rightly claimed

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[1] Cusack (2018), pp. 169–170.
[2] Nietzsche (1996[1887]), pp. 11–38.
[3] Herbrechter (2013), pp. 31–73.
[4] See, Nath (2009), Ch. 1 and 2.

that being 'posthuman' is to attain at least one posthuman capacity. He later explained 'posthuman capacity' as a competence that largely exceeds the maximum attainable horizon of any human being without any assistance of smart technology.[5] His claim indicated toward the current technological advancement, where AI scientists are looking forward to upload the human brain in a supercomputer. The process aims to retain all the memories of a human being forever. Furthermore, it will benefit patients with medical conditions where they can regain their partially/entirely lost memory due to some brain cell damage by the help of this interface. Scientists are looking forward to even transplanting an automated artificial limb which can work as the substitute of a damaged biological organ. As a result, it offers a normal life to any physically challenged person.[6]

Here, posthumanism can help us to clarify the future relations between biological human beings and 'hybrid' posthumans. Here, the fundamental question which needs to be addressed is, 'whether human beings will lose their essence after the amalgamation with intelligent technology and become truly posthuman?'. Furthermore the other question lies here is, 'whether AI technology will be a threat to the biological existence of the most intelligent species in the world?'.[7] We will discuss these issues consecutively in the second and third sections of our paper, whereas the first section contains critical analysis of the genealogical traces of 'posthumanism' and AI technology. However, if we want to find answers to these questions, we should start our journey from the genealogical traces of posthumanistic movement and its influence on the contemporary philosophy. Later, this will help us to explore its' compatibility with AI ethics.

2 From postmodernism to posthumanism

Postmodernism movement had left its consequences on the fields like philosophy, literature, architecture from the mid to late twentieth century. It is the successor of the modernism movement and has taken a discrete stand from the former one. Through the entire era of modernism, society has been habituated with the emergence of a new social, political, and cultural, artistic environment. Contemporary modernists have labelled the traditional structure as outdated and unfit for the modern society; hence, they propagate 'modernism' as the most suitable structure to be followed next.[8] However, in the middle of the twentieth century, a deviating attitude from 'modernism' became significant in the society. It was named as 'postmodernism'. The term 'postmodern' has been first used around the 1870s and has covered different fields like philosophy, art, architecture, drama.[9] The central theme of postmodernism was rejecting the modernistic distinction of high and low art, emphasizing the usage of parody, irony, and criticizing the modernistic lifestyle as a disguise of cruelty. The movement was largely contributed by various eminent French philosophers and literary critiques like Jacques Derrida and many others.[10] Postmodernism has various successors like structuralism, post-structuralism, deconstruction, posthumanism. All these views share a postmodern root, though the central theme is quite different in each of these cases.

'Structuralism' demands that human culture should be understood only through its relation to the broader structure of the society. The analysis of existing structures will help us to make it more effective and understand its authenticity. 'Post-structuralism' has opposed this view by raising counter-argument against various forms of 'structuralism', and it flagged the importance of thinking beyond the prevalent structures. 'Deconstruction' as a type, has been originated by Jacques Derrida.[11] This attitude has been primarily used to understand the relationship between text and meaning. Furthermore, it has a significant impact on contemporary literary theories.[12] It has been used in philosophy to explain the meaning of various concepts and revisit the concepts without any existing radical structures.

The 'posthumanism' movement was another successor of postmodernity. It centered on the 'deconstruction' of the radical anthropocentric hierarchical attitude toward the world. It explicitly rejects the claim that human beings deserve preference over nature or other forms of organisms, because of being the most rational species. As technological advancement has taken a good pace from the late twentieth century, this attitude becomes an inherent part of the postmodernism movement.[13] Along with women empowerment and emphasizing the restoration of natural resources, technological advancement, and the invention of intelligent systems using smart sensors became the trendy outcomes of postmodernism. The progress toward post-humanity gives rise to another conception of 'evolutionary anthropology'.

[5] Bostrom (2008), p. 107.

[6] Zey (2018), pp. 105–116.

[7] Ellul (1964), p. 64.

[8] Lewis (2000), pp. 38–39.

[9] Welsch and Sandbothe (1997), p. 76.

[10] This information is collected from the *Stanford Encyclopedia of Philosophy* of 'Postmodernism', published on February 5, 2015. Aleysworth (2015)

[11] Derrida has expressed his famous theory of '*deconstruction*' in his book *Of Grammatology* (1997). His '*deconstruction*' was a milestone for the analysis of meaning and word. This theory has been used in different literary and philosophical perspectives. It became a fundamental characteristic of the postmodern movement in history. Derrida (1997), pp. 14–24.

[12] This information is collected from the *Encyclopedia Britannica* of 'Deconstruction', published on September 27, 2018. Duignan (2018)

[13] Welsch (2017), pp. 75–86.

'Evolutionary anthropology' has described human beings as worldly beings, which has enhanced themselves through evolution.[14] In contrast, posthumanism preserves the idea of human evolution and tried to extend it further.

The emergence of posthumanism initiates a new framework for conceptualizing 'humanity' among the philosophers. It compels us to think beyond the radical conception of 'humanity' and the urgency to revisit the question of 'whom we can call human?'. It describes the concept of 'human' from the cultural, biological, and technological perspective, using '*deconstruction*' as a method. When 'evolutionary anthropology' came into contact with contemporary modern science and technology, it compelled us to think, 'whether human limitations can be fixed with technology?'. Using smart machinery tools and ushering AI technology has further empowered the posthumanism movement to the next level.[15] The next important questions that strike our mind are, 'what is posthumanism?', 'What are its' practical implications?'.

3 The genealogy of 'posthumanism'

Posthumanism is a conception which has been followed from the postmodernity movement, over the late twentieth century. It is a well-known fact that Ihab Hassan has first critically used this concept in his writing *Prometheus as Performer: Toward a Posthumanist Culture?* (1977). He claimed that we must accept helplessly the rise of post-humanity, because humanism will transform itself into posthumanism, which will be the very end of radical humanism.[16] The central theme of posthumanism considers human beings as the creation of nature which simultaneously coexists with other natural creations (like, other organisms). Humans do not deserve any priority over the existence of other natural creations due to their comparative superior rationality. Posthumanism deconstructs the conception of 'human' and revisits it through the past and present roles played by humans in the world along with other future probabilities. It further includes the hybridization of biological humanity using smart technology and incorporates some traits of 'transhumanism' also.

The concept of 'transhumanism' shares a different root than posthumanism. It is believed that 'transhumanism' has emerged from the '*Enlightenment*' movement in history. According to the genealogical study of the term 'transhumanism', it has been first coined by Dante Alighieri in the fourteenth century. *The Devine Comedy,* He aimed toward the event of biological human beings becoming something 'extra-than-human' in a divine sense. Though there is a touch of ancient mythology and theology in his 'transhuman' conception, it has other significant implications too.[17] Dante has argued in his book *The Divine Comedy* that humans had a *telos*, and their ultimate goal was to be transhuman or posthuman. Here, Dante has portrayed the character of Beatrice, infused with theological and mythical post-humanity.[18] Dante has used 'transhumanism' in a convertible sense with 'posthumanism', as he finds both of these concepts indicate toward exceeding the biological human capacity. William Shakespeare has adopted the same idea in his play *The Tempest*, where he has portrayed Prospero's character with a muse of post-humanity. In the play, Prospero had a magical power to control others' actions and influence them as per his wish. This unearthly magical power turned him into a concrete example of 'posthuman' or 'transhuman'.[19] Over the years, the philosophical posthumanism is thematically related to the deconstruction of radical 'human' conception. In the 1960s, it began as a political cause, became a research project in the 1970s, and in the 1990s developed into an epistemological approach.[20]

The posthumanism movement has rectified the problems concerning anthropocentric attitude towards the world. The movement for preserving nature and its resources is the outcome of it. Nature has its value, and it should not be judged in relation to human benefits.[21] Extinction of any species will be dangerous for the harmony of ecological biodiversity. Posthumanism has influenced the contemporary feminist movement as well. Before the speculation of any feministic thought, it was considered that women are the inferior kind of species, and they do not deserve the same recognition as men in the society. Women were simply denied education, service, and freedom, which was the birthright of men. Through contemporary feminist literature, we came to know that sometimes women were not even treated as human beings, but as an inert property or commodity belongs to men.[22] The feminist movement has taken place in the late nineteenth and early twentieth centuries throughout the entire Western World. The entire period of commotion has been divided into three parts; the first-wave, second-wave, and third-wave feminism movement. The second-wave feminism movement was fueled up by the famous book *The*

[14] Ibid.

[15] Ibid.

[16] Hassan (1977), pp. 830–850.

[17] Pasulka (2018), p. 51.

[18] Alighieri (2003).

[19] Hudson (1888).

[20] Ferrando (2019), p. 2.

[21] This information has been collected from *The Stanford Encyclopedia of Philosophy Archive* (Fall 2017 edition) of 'The Moral Status of Animals', published on August 23, 2017. Gruen (2017)

[22] Beauvoir (2009), pp. 23–38.

Second Sex (1949) by Simone de Beauvoir. Its central theme was achieving equal social recognition, rights, civic facilities, and freedom for both sexes.

The recent advancement in posthumanism movement has been concentrated around the intelligent technological development and the usage of smart systems. In today's world, AI scientists are working toward replicating the human brain in a supercomputer. Philosopher David Chalmers and Andy Clark have argued that our external intelligent gadgets are already proved to be an alternative to our internal biological memory. Therefore, it can be anticipated that in the future, an external super-intelligent technology will be an extension of our whole biological mind.[23] This scenario has been further nourished by the possibilities of 'cyborgs' or 'cybernetic organisms'. Many Hollywood movies like *Star Trek, Terminator, Cyborg*, and web series like *Black Mirror* have flagged the conception of 'cyborgs'.[24] The cyborgs are the result of amalgamation between biological human beings and AI technology. The concept of 'cyborg' is becoming famous among scientists as they anticipate human beings can overcome their biological limitations by this assimilation. This probability has been foreseen by Ray Kurzweil, who has named it as the time of '*singularity*', where our biological limitations will be removed by advanced technology.[25] It will be helpful for genetic engineering, and can even help us to improve our DNA structure. It entails that posthumanism movement has influenced the recent researches in AI field and helps to grow it further.

4 Posthumanism's impact on the field of AI

The field of computer science has conceptualized the field of AI technology in early twentieth century. Subsequently, the field of AI was founded at a workshop held on the campus of Dartmouth College, during the summertime of 1956. John McCarthy was the fore-figure of that workshop, and he further channelized the research progress towards collaborative initiative with Marvin Minsky in 1958.[26] Though the workshop did not come up with any breakthroughs, this is considered as the birthplace of AI technology, because many eminent figures from diverse fields like computer science, engineering, neuro-science have envisaged some good future research initiatives for AI field.[27] The entire period of AI research has received much positive assistance like the participation of thinkers from different fields and government funding in its early days. However, the field has also experienced many downfalls throughout its journey. This period has been named 'AI Winter' in the entire AI research trajectory, which obviously did not last longer.[28] The entire AI field research has received huge funding of nearly 30 years from the US defense agency DARPA (The Defense Advanced Research Progress Agency). As a result, AI scientists have subsequently invented specific goal-oriented intelligent systems which can even beat any human intelligence in a particular field. The invention of IBM Deep Blue was a significant example of what we call 'narrow-AI' (particular goal-oriented intelligent system, which has limited area of expertise). In 1997, the Deep Blue system defeated the world chess champion Garry Kasparov in a chess competition, which became a milestone in AI research history.[29]

As research continues, AI scientists have come up with intelligent sensors, face recognition systems, voice command systems like Alexa, Siri, and many others. In present-day AI systems have been implanted in the medical field; it is even performing the surgery successfully. IBM Watson is currently implemented at Manipal Hospital in the Oncology Department since 2015, and being used as a better predictor than a human doctor.[30] Analysis of big data, usage super-fast processors, and the successful implementation of machine learning have invented a system like IBM Watson. In the present day, the AI field is enriched by brain scientists, neuroscientists, software developers, and experts from biology. All of them are working towards the direction of enhancing human capacity with the help of AI technology in the future and vice-versa. The research has already gifted us the andro-humanoid robots like *Sophia*, developed by Hanson Robotics. The space agencies like NASA and ISRO are replacing human astronauts with humanoid robots for their next space expeditions. ISRO has already planned for its next unmanned *Gaganyaan* space mission with humanoid robot *Vyomamitra*.[31] These humanoid robots are significant examples of posthumans who work as successful replication

[23] Clark and Chalmers (1998), pp. 7–19.

[24] Tegmark (2017), pp. 155–156.

[25] Kurzweil (2005), p. 198.

[26] Russell and Norvig (2010), pp. 17–19.

[27] Ibid.

[28] Bostrom (2014), pp. 6–13.

[29] Kasparov (1996).

[30] The information has been collected from the Healthcare IT News report, which was published on December 3, 2015, by Bernie Monegain (2015)

[31] "Named as *Vyomamitra*, the robot was unveiled at the inaugural session of a symposium organized by the ISRO, International Academy of Astronautics (IAA), and Astronautical Society of India (ASI) on the human spaceflight program in Bengaluru on January 22, 2020. Sivan, during the inaugural function of the three-day symposium titled 'Human Spaceflight and Exploration: Present Challenges and Future Trends' also said that ISRO would undertake two unmanned missions- in December 2020 and June 2021-ahead of India's first manned mission *Gagayaan* in December 2021"- This news was published on http://www.indianexpress.com on January 22, 2020. (Meet Vyom Mitra and ISRO's Robot Which Will Travel to Space 2020)

of human being. *Neuralink Corporation*, an American company founded by Elon Musk in 2016, is working with the aim to create an implantable Brain–Machine Interface (BMI) and enhance biological human brain capacity.[32] It is absolutely a transhumanistic/posthumanistic attitude to the society.

Here, we must clarify that the transhumans or enhanced human beings or cyborgs (mostly used with a symbol of H+ or Humanity Plus) can also be a posthuman, because this conception includes the deconstruction of radical 'humanity'.[33] If we do not go beyond the biological conception of humanity, we shall not be able to call the enhanced human beings or a human being who is dependent on smart technology for his bodily survival as a 'human'. Therefore, it is evident that transhumanism and posthumanism will be embodied in the form of cyborgs or enhanced human beings. The radical conception of 'humanity' exclusively incorporates the biological form of human beings and their biological nature. When we refer to someone as a human being, we could not help us thinking about humanity's biological characteristics.[34] Through the advancement of modern science, technology, and medical science, it has been observed that implementing nano-technological tools are helpful to treat some medical conditions. The usage of 'pacemaker' devices to treat heart issues is an example of human dependence on technology for bodily survival. Here, it can be argued that the implementation of these nano-technical devices in human body can enhance human life span by healing the bodily limitations, proves to be an example of posthumanism and transhumanism. If we articulate humanity only through its biological traits, then any modification or enhancement in biological humanity will be a case of embodied posthumanism.[35]

The hybridization and enhancement of humanity are inherited in the transhumanistic attitude, which has blurred the unsurpassable boundaries between human beings and non-humans.[36] A symposium which has been held at the University of Arts Helsinki in 2017, named as 'Hybrid Matter', has focused on the hybrid ecology, because they anticipate that biological species like human beings, other animals, plants will share a world with AI robots and smart machineries in future.[37] Therefore, it can be claimed that biological beings should also adapt to future changes and enhance their mental boundaries to maintain harmony in society.[38] The fundamental factor of the entire discourse is the relation between posthumans and biological human beings. The exposure of biological human beings' effective interaction with posthumans will be helpful to turn AI beneficial for the society as a whole.

5 The relation between posthuman and human

The posthumans or the cyborgs will be the future companions of our daily life. Many AI scientists and philosophers such as Nick Bostrom, Elon Musk, and Stephen Hawking have already predicted that the AI systems can be a cause of global human extinction if not managed properly. We have seen many movies which have focused on the catastrophic effect of AI takeover on human society. It can be claimed that these movies played a vital part in conceptualizing our future relation between AI robots or posthuman cyborgs and biological human beings. Many movies have already depicted that our human civilization will be in danger if we could not control these technologies. The movie '*I, Robot*' (2004), has portrayed the same issue: there can be a probability of robot malfunction that can be dangerous for human civilization.[39] The film was based on Asimov's three laws of robotics. There are many other movies like *Terminator and Star Trek*, which have pointed to the same issue of AI takeover. However, it can be argued that these are only movies, and cannot be an evident indicator of our future.

The fundamental issue here comes from the question, 'whether human beings will lose their essence after the amalgamation with intelligent technology and become truly posthuman?' The future relation between posthumans or cyborgs and biological human beings will fuel up many ethical issues. First, there will be a paradigmatic shift of radical conception of 'human' as the most intelligent, autonomous and ethical agent. The enhancement of biological human beings, as far as it is possible, might be required, because if we want to achieve a well-balanced future, we should adapt with our technological future.[40] It has been found throughout our human history that the weak are always overpowered by the strong, and the same history can be repeated for the case of posthumans or cyborgs and biological human beings. Posthumanism ascribes subjectivity to other species except

[32] The information has been collected from a blog called '*Wait But Why?*', which has reported about the company's aim with the title of 'Neuralink and the Brain's Magical Future'. The report was published on April 20, 2017, by Tim Urban (2017)

[33] For more information see https://humanityplus.org/about/.

[34] Nayar (2014), pp. 14–18.

[35] Ibid.

[36] Forlano (2017), pp. 16–29.

[37] For more information see http://symposium.hybridmatters.net/. Hybrid Matter's Symposium (2016)

[38] For more information, see http://www.posthumans.org/nyu-symposium-2016.html. Posthuman Future (2016)

[39] This information was collected from 'I, Robot reviews', which is published on *the Metacritic* website.

[40] Bostrom (2014), pp. 283–286.

for humans and goes beyond the boundaries of an anthropocentric view to the world. Even in the present day world, some intelligent machines are far better performers than any human being. Therefore, if posthumans are proved to be a better performer than humans, then it will be a huge blow to anthropocentric superiority as well. If there will be a shortage in scope for human labor, then the posthuman AI robots will be in direct competition with human beings regarding the efficiency to act. Certainly, the former will replace the latter.[41] We can clearly predict the 'posthuman takeover' of the labor industry and artistic powers which will not be a favorable condition for humans. All these leads us to a huge shift from anthropocentric position to posthuman-centric position where posthumans will be the most intelligent species in the world.[42]

Second, as human beings are the programmers or creators of these intelligent systems, we can call those machines as tools for our daily life. So far, they are like a 'servant' or 'slave' to us. However, the 'posthuman-takeover' can replace human creators from their 'masters' position. Here, some AI scientists and philosophers, like Wendell Wallach, have claimed that intelligent technology can be a dangerous master', which will be beyond our control.[43] If we believe the posthumans will be conscious an agent then there lays a fundamental aspect concerning these agents. These posthuman agents will be the eternal 'other' to our human society, independent of the fact that they will be conscious agents or a mere intelligent tool for daily human comforts. Both of these possibilities will lead to a specific situation that those agents can override our rational decisions. Hence, they can be the 'other' to our human society or precisely to human consciousness due to objectification of our consciousness.[44] These agents can easily replicate rational decision-making process and that is why they are assumed to be rational agents so far, who can actively 'think' and 'act'. Therefore, these agents can eradicate humans from the 'masters' position.

Third, the objectification of human consciousness will create a 'master–slave relation' in the mind of biological human beings while dealing with those intelligent systems or posthumans. Hegelian 'Master–Slave Dialectic' can be an expression of our future relationship with those systems due to the objectification of human consciousness by the posthuman consciousness.[45] In support of this claim, Philosopher Mark Coeckelbergh has claimed that "*Whatever its philosophical shortcomings and (undesirable) political luggage, a discussion in terms of the master–slave dialectic can attend us to these problems in human–technology and human–human relations, including alienation and knowledge, but also for instance power issues and the issue of automation and (un)employment*".[46] Therefore, it can be claimed that these posthumans have the power to takeover on our cultural and artistic aspect and can pose a threat to human '*bildung*' or artistic agency. In this context, Philosopher Joana J. Bryson has suggested that robots should be treated as slaves, and we should not humanize them or consider them as moral agents, but rather as tools for our daily comfort.[47] If these posthuman agents go beyond our control then we will be enslaved by our own creation and it will be a serious ethical disaster. Therefore, it is evident that 'posthuman-takeover' can develop a conflict is our existing normative structure as well.

Fourth, the 'weak AI' supporters have already propagated the view that AI robots can be a simulation of conscious activities, but they might not achieve actual consciousness. For them, consciousness is a biological feature, which cannot be generated artificially, and they prefer an AI system to be manufactured for specific and limited goal-oriented task. They further claimed that as these intelligent agents will perform under the supervision of humans, the unforeseen problem overpowering by 'strong AI' systems will no longer exist.[48] This whole argument leads the field of AI to shift its emphasis from 'strong AI' to 'weak AI'.[49] The above analysis indicates towards the change of the power structure between humans and posthumans. As posthumans will be the most intelligent species in comparison to humans. Investors might invest on the future research of smart technology more to increase their profit level.[50] Though here, we have only considered the 'weak AI' robots so far, still there is a possibility that those 'powerful nations' or 'powerful persons' can use AI technology in a biased way, and that might cause severe damage to our future world. Therefore, our whole power structure will be redefined along with future human–posthuman relation.

Our future research on AI should be meticulous about all these possibilities of conflicts in order to manufacture a beneficial AI. As we consider the plausibility of posthumans, we must evaluate the ethical concerns related to this issue. Obviously, the future relation of humans–posthumans will be a fundamental factor in deciding whether those posthumans deserve any exclusive rights, and recognition will simply serve the needs of human beings. It will be essential

[41] Tegmark (2017), pp. 114–128.

[42] Ibid.

[43] Wallach (2015).

[44] Nath (2017), pp. 563–571.

[45] Hegel (2018[1807]), pp. 108–116.

[46] Coeckelbergh (2015), p. 229.

[47] Bryson and Wilks (2010), pp. 63–74.

[48] Bostrom (2014), pp. 140–148.

[49] Searle (1980), pp. 417–424.

[50] Bostrom (2014), pp. 4–5.

to decide the code of conduct for those posthuman AI robots, and subsequently, we could decide whom to prioritize in preference. The responsibility question remains the most debatable issue in this regard and its ethical implications. Thus, in the next section, we shall discuss the importance of AI ethics from the outset of posthumanism.

6 Why is 'ethics of AI' important?

We are moving towards a future world with these technologies, and we have to interact with cyborgs/posthumans robots in our daily life. That means we have to take care of the ethical dimension of AI systems in relation to humans. AI systems have been invented with an idea of creating autonomous machines which can ease our everyday life. However, over time, the goal has been shifted to invent an autonomous ethical agent, which can act like human beings without any human intervention for betterment of our society. Susan Leigh Anderson and Michael Anderson have summarized this goal in their writing, "*Ideally, we would like to be able to trust autonomous machines to make correct ethical decisions on their own, and this requires that we create an ethic for machines*".[51]

Previously, we discussed that 'posthuman-takeover' can be a threat to the humans. That means, we have to face the question, 'whether AI technology will be a threat to the biological existence of the most intelligent species of the world?'. There are different possibilities which can be argued from the ethical outset of humans. First, generally, the term 'ethical' applied to those who have the power to judge actions as agreeable or not, which means the term is solely applicable to human beings only, as we believe that only human beings are entitled with reason to judge something. It seems unnatural to call any artefacts as ethical or unethical in the same sense.[52] Because, 'thinking' is considered as a 'conscious action' which requires a prior 'conscious agency'. The person who has a conscious agency can be called as the agent of that particular action. Therefore, it can be claimed that being an agent of action presupposes the consciousness of performing that action in a particular space and time. However, one can raise the issue that AI robots are not like a conscious human being who can be aware of their performance. They are simply an intelligent programmed system capable enough to accomplish the task assigned to it by its user. Here many questions arise 'whether these posthuman agents will be competent enough to perform any moral action?', 'If these agents go immoral then who should be responsible for it? Is it the machine, the programmer, the manufacturer company or the end user?'. The AI scientists are still in a dilemma if they could ever be able to create a conscious machine, or it will be a simple machine that can only follow the command to mimic the moral action. The answer to this question is entirely dependent on the future invention of the AI field.[53]

Second, apart from the question of 'consciousness', it can be further claimed that artificial robots' actions will impact our civilization in both positive and negative ways. Therefore, the AI field has the potential urge to shift their focal point from accuracy of performance to 'moral agency' or 'agency to hold the responsibility'. Without any 'conscious moral agency', we could not held them responsible for their actions. Rather the whole responsibility will be on human participants who were part of their creation. Therefore, the question of responsibility will be an ethical issue in AI ethics. As all the participants are not aware of these course of actions or 'whether or not any intelligent system is actually implemented?'(like, many social media user are not aware of AI algorithms implementation). That is to say, we cannot hold these participants responsible for these systems actions.

Third, in relation to the above point, we adhere that 'consciousness' and 'thinking' hold a fundamental role in ethical performances. Here, 'thinking thing' implies the 'subjective thinking thing', because a subject is required for rational 'thinking'. These subjective behaviors are non-mechanical due to its' non-corporeal features. These conscious practices are free intentional actions of a 'thinking subject'. In this regard, we must assert that Descartes' concept of 'I think' presupposes subjective experience, because it is 'I' who experiences the world. Here, Descartes' notion of 'I' contradicts the posthumanistic conception of mind. The essence of mind is 'thought', and the 'act of thought' is identified with the 'act of consciousness'. Therefore, it follows that cognitive actions are conscious engagements, rather a mere mechanical process. Thus, for Descartes, one of the most important aspects of cognitive states and processes is their phenomenality. Because human rational capacities can be defined only in relation to their conscious 'agency', not to their posthumanistic 'body' so far. Descartes' dictum, "I think, therefore, I am" not only establishes the existence of the 'self' which is responsible for 'thinking' and 'rational actions', but also freed itself from bodily mechanistic laws.[54] Hence, an idea of ontology has confined the idea of 'self' from the notion of 'posthumans' and raises a question of ontology for the later. As our aim in this paper is to analyze the scope of morality in posthuman terrain, we will follow the same path keeping 'the question of ontology' aside.

[51] Anderson and Anderson (2011), p. 1.

[52] Wallach and Allen (2009), p. 55.

[53] Nath and Sahu (2020), pp. 103–111.

[54] Descartes (1984), p. 17.

Fourth, as far as the moral decision making of robots is concerned, Isaac Asimov had introduced three basic rules for robots to be moral agents. However, that is not sufficient enough for the future, as we have travelled a long path from the primitive stage of creating AI.[55] Nowadays, our daily life is designed by many automated systems, which is capable enough to replicate rational and intelligent decision making. Even AI scientists are anticipating that shortly a superhuman artificial general intelligence(AGI) can be invented, which can control its actions and command feeding(programming) on its own. That means, heuristic AI agents can learn from their own experiences and it can even accommodate biases from their databases. It indicated that if it goes wrong or malfunctions in any way, it will certainly turn up as a devil for the entire civilization.[56] Therefore, we need a more sophisticated ethical structure for AI agents from the very beginning, to turn it as a faithful companion for our society. To get a proper solution to this conflict, we have to analyze whether the term 'moral agent' is applicable to the AI agents in the same way it applies to us or is there an implicit minute difference in them.

Fifth, when we call a human being a moral agent, we assume that the performer has complete knowledge of his performance; therefore, he will be held responsible for his action's impact. The present status of AI robots is, they are still dependent on the initial command feeding method. That indicates these agents are not aware of their actions like humans but these actions will undoubtedly impact human civilization in all possible ways.[57] In other sense, we can claim that future transhuman AI agents should incorporate some set of rules to avoid any hazardous bias. If we assume them as a future companion of our daily life, then indeed, their actions require a guiding framework to make ethical choices of actions. It can be derived from this situation that the programmer must have some specific human value in his mind, at the time of programming that intelligent machine. That specific human value here is *happiness or pleasure*.[58] However, these happiness principles sometimes emerges as a conflicting case when we need to choose between two similarly important outcomes.

Sixth, it can be claimed that AI robots will be better performers than any human beings, because human feelings have often deviate from their goals, which might not be valid for the AI robots or cyborgs. They will be more objective performers than any human being, because they will be devoid of human emotions, which we consider as entirely biological. However, their initial set of normative rules (like, do not harm any human being) will not be sufficient to convert these posthuman agents as a genuine moral agent. We should feed them with the knowledge of the human value system to make them beneficial for the betterment of human society. The implementation of human values in the AI system is not an easy task to perform. Nick Bostrom has mentioned in this same context, "Goal system engineering is not yet an established discipline. It is not currently known how to transfer human values to a digital computer, even given human-level machine intelligence".[59] Therefore, it is believed that there should be always a scope for humans to override these machines decisions on a certain point of time (though it might not be feasible in certain real-life scenarios, like, the self-driving car).

It can be derived that AI ethics will be an inherent part of AI implementation in our society, and it will be dependent on future development of AI field. Posthumanism's effect on the ethics of AI will play a significant part in forming the required ethical structure of AI implementation. If AI scientists are ever successful in inventing the AGI system in the future, that will be a significant example of an enhanced hybrid-human being or a posthuman/transhuman. It might be possible that those artificial posthumans will deal with our ethical dilemmas differently, and their solution might not be conceivable to us, because those systems will be far more intelligent than us.[60] The way they will judge the ethical dilemmas might require a more intellectual level than we have. However, posthumanism left its impact on our normative domain and redefined the dimensions of 'moral' agency. In the next section we will continue to deal with this issue.

7 Posthumanism and ethics

The philosophical posthumanism movement has redefined the relation between a non-human entity and human beings. Posthumanism has affected not only the field of AI but also the emergence of ethics of AI. As far as posthumanism is concerned, there is a difference between human-nature relations and human–technology relations.[61] The former incorporates both biological entities, whereas the latter incorporates one biological and another constructed artefact. The philosophical posthumanism will hold different approaches to address both of the cases. Those approaches will be a crucial factor in the creation of ethical AI. As the ethics of AI is being visualized as the code of conduct for the AI robots, it is primarily focusing on the factor that if these AI robots

[55] Asimov (1950).

[56] Elon Musk has claimed that "With artificial intelligence, we are summoning the demon", MIT. McFarland (2014)

[57] Kurzweil (2005), pp. 16–36.

[58] Bostrom (2014), p. 227.

[59] Ibid, p. 253.

[60] Ibid, pp. 266–270.

[61] Jones (2018), p. 365.

will be helpful for humans or will be harmful to our societal development. Posthumanism teaches us that humans are not at the centre of ethics. Other factors and species are similarly valuable and hold their own position in ethical situation.

In contrast, till date we are focusing on human interest whenever we are trying to articulate the ethical code of conduct for AI robots. Therefore, it is a conflicting situation from the outset of posthumanism. If we consider the cyborgs or AI robots as posthumans, then our contemporary posthumanism perspective will compel us to give the same importance to the interest of those future cyborgs and posthumans. Here, a scope of robot rights is bubbling up on the surface, and if we consider that AI robots or cyborgs will be more intelligent beings that we are, then certainly those intelligent beings deserve some rights like our fundamental rights. If posthumanism's impact on the ethics of the AI field invokes robot rights in the future, then a conflict between human rights and robot rights will be possible.[62]

It is practically possible that in a specific situation, human rights are coming in conflict with the robot rights, and then it will be difficult to choose the most ethical decision. If posthumans get the preference, that will certainly not be acceptable to biological human beings. On the other hand, if we prioritize human rights over robot rights, then it can be claimed that the ethics of AI will be redundant as it will ultimately return to the anthropocentric attitude. Therefore, the problem of robot rights will be a future problem for both the posthumanist field and the ethics of the AI field. The conflict between the ethical fields of human beings and posthumans can only be solved when posthumanism will eventually further enhance the future scope of ethical domain. As posthumanism's central theme is enhancing the boundaries beyond radical humanism, it can improve the ethical dilemmatic condition by enhancing the boundary of ethics itself. We should welcome the redefined ethical solution proposed by AI agents or posthumans and it will be helpful to explore new dimensions of rationality and intelligence.[63] Therefore, the future value system can be formed with an enhanced scope, which will include both biological human beings and the posthumans.[64] This extended field can even include the natural resources and other species as well, and the core value for this realm should be 'having the opportunity to explore the transhuman and posthuman realm'.[65]

Another impact of Posthumanism is it points towards scientific objectivity rather than humanistic subjectivity in ethical domain. Humans judge all the actions through subjective factors like emotions, feelings. However, the posthumans are supposed to be free from these types of subjective biases. The problem is when both of the subjective factors and objectivity become incompatible with each other. In dealing with ethical issues, we do not always follow the objectivity ground, and sometimes we judge the issues from a compassionate ground of humanity.[66] However, the posthumans will acquire more sense of objectivity like intelligent machines, and they will be prone to judge ethical issues with a sense of objectivity. Their ethical judgment may be coming into conflict with the biological human being's ethical judgement. Hence, there should be particular guidelines in the future posthumanistic society, which will incorporate both the humanistic factors as well as the posthumanistic subjectivity. Otherwise, there will be chaos in the judicial procedures of ethical issues.

We can conclude from these above arguments that posthuman ethics and human ethics can face practical conflicting situations if followed the radical boundaries. Here, the fundamental issue is 'whether or not we should ascribe full 'moral agency' to our future posthuman agents concerning the fatalities of their malfunctions and overriding human control?'. Their 'moral agency' has been termed as 'functional morality' rather than a 'responsible moral agency'.[67] As these agents are not answerable and do not possess a sense of responsibility, their 'moral agency' is considered as an imitation of human moral agency. However, if we maintain this position then humans will be at the top of the genuine 'moral agency'. Therefore, anthropocentric preference will be inevitable in that circumstances, which will turn the posthumanistic root redundant. That means, the return of 'anthropocentrism' in relation to 'AI ethics' will be an important factor to be evaluated further. In our next section, we will critically evaluate this issue.

8 Will 'ethics of AI' be a return of 'anthropocentrism'?

The central theme of AI ethics is to produce a '*friendly-AI*', which will perform its duties for the betterment of human society. Now, the question is 'when we can call an AI system as '*friendly AI*'?. The term '*friendly AI*' indicates a human benefiting and non-human harming AI system, which can be used to accomplish real-life goals and improve our future days.[68] In other words, it can be claimed that a '*friendly AI*' will be an ethical, moral agent to us, as those systems will work according to our ethical values. Now, here a conflict emerges from the humanistic analysis of what is moral to

[62] Gunkel (2018), pp. 87–99.

[63] Coeckelbergh (2020), pp. 34.

[64] Bostrom (2005), pp. 87–101.

[65] Ibid.

[66] Chiew (2014), pp. 51–69.

[67] Wallach and Allen (2009), p. 39.

[68] Yudkowsky (2011), p. 2.

us. We are prone to judge future intelligent systems from the anthropocentric attitude, which we should be careful of. It can be further concluded with a question of 'whether posthumanism will finally finish its journey in the boundaries of anthropocentrism or it will explore a diverse dimension free from our radicalistic perspective?'. This analysis can be further extended with a claim that the ethics of AI presupposes the complete knowledge of human society and its culture. Otherwise, it will not be relevant to contemporary society. It has been proved that rules which were present in ancient societies have modified themselves as per the requirements of the present society and its cultural structure. Therefore, it can be construed that if AI takeover is an expression of posthumanism, it also demands the comeback of structuralism as far as its implementation in society is concerned.

If ethics can be implemented in artificial systems, it will be a revolutionary invention for the entire humankind. Our everyday life will be more relaxed, and we can embrace AI agents as our day to day companion. Even those systems can be used fruitfully in such hazardous fields where human life could be at stake while working. Therefore, all of us can anticipate positively that on one beautiful day, scientists will be able to come up with the ever-awaited super-intelligent artificial moral agent as a faithful companion of our future world. However, the question remains 'whether super-intelligent AI robots will follow the human value structure or innovate a new value structure that can accommodate both the human and posthumans?'.

It can be construed that the implementation of AI systems in our society demands some ethical guiding framework so that those systems will not turn to be worse for the entire civilization. Now structuring an ethical framework is not an easy job for the programmers of the AI system. We could raise our concern that without the 'subjective' feeling of an agent, which is precisely a presence of a conscious mind, acting morally is not possible. Therefore, the consciousness of AI robots will be an open question for the entire field of technology. Furthermore, a chance of implementing subjective bias in those AI systems will be a problem for the field of AI. An automated and smart system will undoubtedly need initial programming of specific rules. Otherwise, they can collect biased data from their surroundings. After the initial guidance, those systems may be able to work according to their experiences and analysis of big data. That is why many AI scientists have already proposed for the fusion of both '*top-down*' and '*bottom-up*' approaches to construct the right ethical framework for AI agents.[69] However, the proposed fusion is not vastly accepted among all the AI scientists. Attempting to create a moral or compassionate machine from the outset is more likely to result in an ethical expert system rather than in a self-aware being.

Inclusion of ethics in the programming level of autonomous systems may be possible by brain simulation, or it may be programmed as a command system in it. In any of these approaches, there will be different challenges. One of the fundamental problems of AI ethics will be making ethics computable by creating a program that enables a machine to act as an ethical advisor to human beings. However, we should anticipate that our moral theories could be formatted in a computable version entirely, and they will easily be implemented in AI systems in the near future. In this context, we can conclude that if the formatting of moral values will be possible soon, those commands will be in the form of 'hypothetical conditions' of an action.[70]

From the above discussion, it can be derived that no one has invented any comprehensive approaches for ethical implementation in AI robots until today. Many philosophers and scientists are currently dealing with this issue, and sometimes they come up with a view that implementing ethics in an AI system will not be possible. For them, we should provide some initial guiding framework for those systems, after gaining a certain amount of intelligence, those systems will learn how to act ethically on their own from their data sets and analysis of the data. They are a supporter of the '*bottom-up*' approach to ethical implementation. Some philosophers have proposed that collaboration of all these ethical theories will be the most favorable approach for ethical implementation, as any one of these theories will not be sufficient for the system. They are precisely the supporters of the '*top-down*' approach to ethical implementation.[71] As the intelligence level of today's AI systems is still at an elementary level in comparison to its future expectations, we could not be able to provide the ultimate ethical approach for it and can only anticipate the future based on our current understanding of it. The '*top-down*' or '*bottom-up*' approach is entirely related to the existing human value alignment system and ethical theories. For a future society which will incorporate posthuman cyborgs and AI robots will not find this value system sufficient. Human ethical values should modify itself according to the changes in society, and it will be better if it extends its boundaries and incorporates the posthumans too.

9 Conclusion

It can be visualized that AI robots or enhanced cyborgs will be our future fellow companions, and this technological advancement will change our societal value structure. It can be construed that posthumanism is the birthplace of

[69] Etzioni and Etzioni (2017), pp. 403–418.

[70] Powers (2009), pp. 15–16.

[71] Etzioni and Etzioni (2017), pp. 403–418.

the technological infusion in the radical concept of humanity and modifying the same. In the future, our conception of humans will also be changed as biotechnology, genetic engineering, stem cells, and cloning will lead us to a different conception of humanity. As of now, we are claiming human beings as the driving force of our technology, but this reality can be changed in the future.[72] It is entirely plausible that smart technology will be the driving force of the future, and our lives will be dependent on the actions of the posthuman AI robots or cyborgs. Because human society is always at risk from some privileged group, and they entrap other groups of people in their slavery. If that is the tradition followed by human society through the entire historical trajectory, then the same tradition can be repeated for the case of genetically or technically enhanced people and rational biological human beings.[73] Therefore, our future society should be formed in such a way that it can take care of both the privileged and biological members too.

Genetical engineering, stem cells, all these technologies aim towards immortality and go beyond humans' natural limitations.[74] Death is considered a natural process, and it will help to maintain the balance in our ecology and biodiversity. If technological advancement is advanced enough to make death as per our choice, it can also be harmful to the ecological balance. As the death of older people maintains control over our population, it would be challenging to control the population explosion throughout the world if technological advancement will confer us with immortality. Therefore, modifying the natural world can be extremely dangerous and will imply a catastrophic effect on the entire world.

In the present time, we are using AI technology as mere tools to achieve our goal with ease. However, if the superintelligent agent is a part of our future world, it will be possible to replicate consciousness artificially. We believe that 'consciousness' and 'thinking' are the essential characteristics of 'humanism'. The last one is only a property of the most intelligent living being of the world. However, sometimes people start to think the AI agents as their true companions and develop a sense of empathy towards them. The current development of AI is still in a phase where scientists are in double mind whether they can ever achieve a thinking machine or will always be a rule-based performer in the future. If we are anticipating to confer AI machines with a thinking capacity, then the so-called posthumanism will take a step towards another conflict. We need enhanced 'wisdom' and 'intelligence' to solve the future ethical problems concerning posthumans as the peers. AI or any smart technology can help us to analyze the patterns to reach the best decision, but certainly these machines cannot be crowned with 'wisdom'. As far as the genuine 'moral agency' is concerned, we cannot hold these posthuman agents fit in this structure. We need to develop a good adaptation with future technologies to make it beneficial for the entire world.

[72] Ferrando (2019), p. 255.

[73] Bostrom (2003), p. 33.

[74] Francis Fukuyama has demonstrated the negative impacts of genetic engineering in his book *Our Posthuman Future: Consequences of The Biotechnology Revolution* (2002). He demanded strict governance for its application to protect it from any kind of misuse. pp. 181–218.

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